

Statistical methods appendix

China-US schizophrenia burden study

Provisional methods appendix: official GBD 2023 population and registered NCI Joinpoint outputs remain required before submission.

Analysis estimands

- All-age numbers describe service-volume burden.
- Age-standardized rates describe temporal and cross-country rate patterns after standardization.
- Derived ratios and changes are point estimands; no draw-based uncertainty is asserted.
- Decomposition components are deterministic functions of posterior mean rates and populations.

Segmented log-linear trend model

For year t and rate $r(t)$, $\log r(t) = \beta_0 + \beta_1(t-t_0) + \sum[j] \delta_j \max(0, t-\tau_j) + \text{error}(t)$. Candidate models contain zero, one, or two knots and require four observations between boundaries. Sequential residual-permutation tests compare zero versus one and one versus two knots. Segment APC is $100[\exp(\text{slope})-1]$; AAPC is the time-weighted mean log slope transformed to a percentage.

The provisional build uses 4,499 permutations and random seed 20230707. It is an independent implementation, not NCI Joinpoint. Official NCI inputs and settings are exported separately.

Parallelism tests

For each prespecified pair, a common spline basis used the union of selected knots. The restricted model allowed a group intercept but common slopes; the full model added group-by-slope interactions. An F test compared residual sums of squares. Benjamini-Hochberg correction was applied within the country-comparison and sex-comparison families.

Shapley decomposition

For total adult population N , age-share vector S , and age-specific rate vector R , reconstructed burden is $B = N \sum_a (S_a R_a)$. Each factor was replaced from baseline to endpoint in all $3!$ orders. A factor's contribution is the average marginal change across orders. The three contributions therefore sum exactly to $B_{\text{end}} - B_{\text{start}}$ apart from floating-point tolerance.

APC estimable functions

The exact relation cohort=period-age prevents simultaneous identification of unrestricted linear age, period, and cohort effects. The model therefore contains an intercept, identifiable longitudinal age slope, net drift, and nonlinear age, period, and cohort bases constrained to be orthogonal to intercept and linear trend. Net and local drift are estimated separately from annual log-rate regressions. Reported relative risks are curvature contrasts to central reference categories.

Uncertainty taxonomy

Quantity	Reported uncertainty	Interpretation
Original GBD estimate	Native 95% UI	2.5th-97.5th percentiles of GBD draws
Segment APC/AAPC	Model 95% CI	Conditional regression uncertainty; not GBD posterior uncertainty
Ratios and changes	None	Point estimate; interval dominance is conservative only
Decomposition component	None	Deterministic attribution using posterior means